

with any form of tag, including an anonymous tag. The tags within a list do not need to be unique with respect to other members of the list.

A.6. Element Encoding

A TLV element is encoded a single control octet, followed by a sequence of tag, length and value octets. Depending on the nature of the element, any of the tag, length or value fields may be omitted.

Control Octet	Tag	Length	Value
1 octet	0 to 8 octets	0 to 8 octets	Variable

A.7. Control Octet Encoding

The control octet specifies the type of a TLV element and how its tag, length and value fields are encoded. The control octet consists of two subfields: an *element type field* which occupies the lower 5 bits, and a *tag control field* which occupies the upper 3 bits.

A.7.1. Element Type Field

The element type field encodes the element's type as well as how the corresponding length and value fields are encoded. In the case of Booleans and the **Null** value, the element type field also encodes the value itself.

Control Octet								Description
Tag Control			Element Type					
7	6	5	4	3	2	1	0	
x	x	x	0	0	0	0	0	Signed Integer, 1-octet value
x	x	x	0	0	0	0	1	Signed Integer, 2-octet value
x	x	x	0	0	0	1	0	Signed Integer, 4-octet value
x	x	x	0	0	0	1	1	Signed Integer, 8-octet value
x	x	x	0	0	1	0	0	Unsigned Integer, 1-octet value
x	x	x	0	0	1	0	1	Unsigned Integer, 2-octet value
x	x	x	0	0	1	1	0	Unsigned Integer, 4-octet value
x	x	x	0	0	1	1	1	Unsigned Integer, 8-octet value
x	x	x	0	1	0	0	0	Boolean False
x	x	x	0	1	0	0	1	Boolean True
x	x	x	0	1	0	1	0	Floating Point Number, 4-octet value
x	x	x	0	1	0	1	1	Floating Point Number, 8-octet value
x	x	x	0	1	1	0	0	UTF-8 String, 1-octet length